Lecture II: Explore & Exploit

Programming: Everyday Decision-Making Algorithms

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Today in one sentence

Whether to try something new or stick with what you know depends mostly on how much time you have left.

WHY IT MATTERS

You have 45 minutes for lunch: the new restaurant or your reliable favorite? Companies face the same choice every day: a new supplier or the proven one, a new marketing channel or the one that works, a new film or another sequel.

Roadmap

01 The dilemma: try something new or stick with what works

02 Time decides: why the end of the interval changes everything

03 Smart strategies: from rules of thumb to the Gittins index



You have 45 minutes for lunch. Do you try the new restaurant or go to your reliable favorite?

Quick Poll: Raise your hand, who would try the new restaurant?

Speaker notes

This decision happens hundreds of times daily, and there’s optimal math behind it.

Explore gathers information, exploit uses it

Explore

- **Definition:** Gathering new information
- **Risk:** Unknown outcomes
- **Example:** Trying that new sushi place

Exploit

- **Definition:** Using known information
- **Risk:** Missing better options
- **Example:** Your go-to pizza joint

> In any given moment you can only do **one of the two**.

Speaker notes

The fundamental trade-off: explore and exploit are mutually exclusive choices in any given moment. You can’t have your cake and eat it too. Exploring means gathering new information with unknown outcomes, for example trying that new sushi place. Exploiting means using known information, with the risk of missing better options, for example your go-to pizza joint.

Examples from Your Life

Question: Where do you face explore vs. exploit decisions?

Explore or exploit: three examples

Medical

Explore: Test a new drug on patients.

Exploit: Use the proven treatment.

Tech

Explore: Test a new website design.

Exploit: Keep the current design.

Dating

Explore: Meet someone new.

Exploit: See someone you already know.

Question: What is the optimal balance?

Speaker notes

Exploring brings information at a price: unknown side effects, development time and lost conversions, bad dates. Exploiting brings a predictable result at the risk of missing something better. The million dollar question: what is the optimal balance between exploring and exploiting? Science has been asking this for over 50 years.

What if you only exploit?

Question: What happens then?

- Exploitation is **not always the best strategy**
- Especially with **very limited information**
- If you **stop exploring**, you **miss better options**

What if you only explore?

Question: What happens then?

- Exploration is **not always the best strategy**
- If you **constantly explore**, you **never enjoy the results**
- Imagine eating each meal, hearing each song, meeting each person **only once**

Computer scientists call it the multi-armed bandit

- Computer scientists formalized the explore-exploit trade-off
- It is known as the **multi-armed bandit problem**
- **Question:** What is a one-armed bandit?

One Armed Bandit



Three slot machines, 20 pulls: which do you play?

Imagine: You walk into a casino with **3 slot machines**

- Each machine has a **different** (unknown) win rate
- You have **$100** and **20 pulls** total
- **Goal:** Maximize your winnings

- **Explore:** try different machines. **Exploit:** play the best one so far

Speaker notes

How is this related to explore vs. exploit? Exploring means trying different machines to learn their win rates; exploiting means playing the machine that has performed best so far.

Every pull costs money and gives information

- Each pull tells us more about a machine's chance of winning
- But each pull also **costs** money

Question: How do you decide which machine to play next?

Interactive Casino Challenge

You've been playing for a while. Here's what you know:

MACHINE	PULLS	WINS	WIN RATE	YOUR CHOICE?
A	15	9	60%	
B	2	1	50%	
C	0	0	???	

Class Vote: A (known performer), B (limited data) or C (complete unknown)?

Follow-up: What if you only had 2 pulls left? Or 100?

Machine A looks better, but both estimates are shaky

- Expected value = average wins per pull
- E(A) = 9/15 = 0.6 and E(B) = 1/2 = 0.5
- But 2 and even 15 pulls are **few observations**

Speaker notes

The expected value is the average number of wins per pull. Machine A has the higher expected value, but with so few pulls the estimates are uncertain: machine B could easily be the better machine.

The best choice now is not always the best choice overall

- The bandit problem stands for many real-world decisions
- The best choice **now** can differ from the best choice **in the long run**
- Which lever to pull next depends on one more thing:

Roadmap

01 The dilemma: try something new or stick with what works

02 **Time decides: why the end of the interval changes everything**

03 Smart strategies: from rules of thumb to the Gittins index

How long will you stay in the casino?

- **Question:** Why does this matter?
- **Question:** How does it change our choice between machine A and B?

> “I’m more likely to try a new restaurant when I move to a city than when I’m leaving it” (Chris Stucchio)

Long interval: explore. Short interval: exploit.

- The time you plan to stay in the casino is **the interval**
- **Long interval:** explore more, you have time to use what you learn
- **Short interval:** exploit what you already know

Speaker notes

The optimal strategy depends on the length of the interval. This is the single most important idea of today: the same decision can be right at the start of an interval and wrong at its end.

Explore while you still have time to use what you learn

> “I moved to Pune, India, and I just [...] eat everywhere that didn’t look like it was gonna kill me” (Chris Stucchio)

Exploration



Exploit when you are ready to cash in

> “And as I was leaving the city I went back to all my old favorites, rather than trying out new stuff [...]. Even if I find a slightly better place, I’m only going to go there once or twice, so why take the risk?” (Chris Stucchio)

Exploitation



Hollywood exploits: 9 of the top 10 films are sequels

Sequels among the 10 highest-grossing films

YEAR	NUMBER OF SEQUELS
1981	2
1991	3
2001	5
2011	8
2024	9[1]

Question: What explains this trend, and what does it say about Hollywood's interval?

1. Box Office Mojo, top 10 by US domestic gross 2024 (boxofficemojo.com/year/2024). Counted as sequels: all except *Wicked*; *Twisters* counted as a sequel.

Speaker notes

Class discussion: What explains this trend? How does this relate to explore vs. exploit? What does this suggest about Hollywood’s time horizon?

Sequels...



Studios behave as if their interval is ending

- A new movie is risky, but can create a new fan base (**explore**)
- A sequel has a guaranteed fan base, a sure thing (**exploit**)
- Studios may believe they are near the **end of their interval**
- So they pull the best arms before the casino turns them out

Anecdotes are not a strategy

- Anecdotes help us understand the trade-off, but they are no solution
- An exact algorithm for the trade-off is **very hard** to find
- On the way, there were many interesting approaches...

Roadmap

01 The dilemma: try something new or stick with what works

02 Time decides: why the end of the interval changes everything

03 **Smart strategies: from rules of thumb to the Gittins index**

Rule of thumb: win-stay, lose-shift[1]

- **Question:** What do you think this strategy does?
- **Won?** Stay with the same arm. **Lost?** Switch to another arm.
- Simple, and proven better than choosing an arm at random

1. For more details see Robbins, H. (1952) 'Some aspects of the sequential design of experiments', Bulletin of the American Mathematical Society, 58.

Win-stay makes sense, lose-shift is too hasty

Win-stay

- If you win, pull the same arm again
- The arm has just become more attractive

Lose-shift

- Switching after every loss is a rash move
- Your favorite restaurant, tenth visit: always great, today disappointing
- Should you never go back?

Easy answers come with problems

- Win-stay lose-shift **penalizes losses too much**
- It ignores **the interval**
- But it was a good start for better approaches

A fair estimate: Laplace's rule of succession

- First we need a fair estimate of each machine
- The plain win rate is extreme after few pulls: 1 win out of 1 pull looks like 100%, 0 out of 0 is undefined
- Laplace (18th century) suggested a fairer estimate: after w wins and l losses, the chance of a win is about

$$\mathbb{E}[\mathbf{win}] = \frac{w+1}{w+l+2}$$

- In words: add one imagined win and one imagined loss, then take the share of wins

Speaker notes

Pierre-Simon Laplace suggested this rule in the 18th century. The two added pulls keep the estimate away from 0 and 1 when we have seen very few pulls. The rule estimates how good a machine is; it does not tell us when to stop exploring.

Try it: machine A and B by Laplace

$$\mathbb{E}[\text{win}] = \frac{w+1}{w+l+2}$$

- **Question:** What is E[B]?
- **Question:** What is E[A]?
- **Question:** Which machine looks better by this estimate?

- E[B] = (1 + 1) / (2 + 2) = 0.50
- E[A] = (9 + 1) / (15 + 2) = 10/17 ≈ 0.59
- E[C] = (0 + 1) / (0 + 2) = 0.50, even though C was never played

Laplace gives unknown options a fair chance and avoids estimates of 0% or 100% after a few pulls.

Laplace estimates for wins and losses

LOSSES/WINS	1	2	3	4	5	6	7	8	9	10
1	0.50	0.60	0.67	0.71	0.75	0.78	0.80	0.82	0.83	0.85
2	0.40	0.50	0.57	0.63	0.67	0.70	0.73	0.75	0.77	0.79
3	0.33	0.43	0.50	0.56	0.60	0.64	0.67	0.69	0.71	0.73
4	0.29	0.38	0.44	0.50	0.55	0.58	0.62	0.64	0.67	0.69
5	0.25	0.33	0.40	0.45	0.50	0.54	0.57	0.60	0.63	0.65
6	0.22	0.30	0.36	0.42	0.46	0.50	0.53	0.56	0.59	0.61
7	0.20	0.27	0.33	0.38	0.43	0.47	0.50	0.53	0.56	0.58
8	0.18	0.25	0.31	0.36	0.40	0.44	0.47	0.50	0.53	0.55
9	0.17	0.23	0.29	0.33	0.38	0.41	0.44	0.47	0.50	0.52
10	0.15	0.21	0.27	0.31	0.35	0.39	0.42	0.45	0.48	0.50

Bellman: calculate backwards from the last pull

- Richard Bellman found the exact solution for a **known interval**
- Start at the **final pull**: with no future left, simply exploit
- Then calculate **backwards** the best choice in every earlier situation

Speaker notes

A few years after win-stay lose-shift, Richard Bellman found an exact solution for all cases with finite and known intervals. The question is when to stop exploring and start exploiting. The solution uses dynamic programming: start at the final pull and calculate backwards what the best choice is in every possible situation. This works because the last pull is easy.

Bellman needs to know how long you stay

- The calculation grows fast with many arms and a long interval
- Worse: we usually **don’t know** how long we stay in the casino
- So the bandit problem seemed to have no practical solution

Speaker notes

The calculation of the optimal strategy is very extensive when there are many arms and a long interval. And the approach does not help in many scenarios, because we do not know the exact length of the interval. At this point it looked like the multi-armed bandit problem would remain a problem without a solution.

Gittins: a drug-trial problem at Unilever[1]

- In the 1970s, Unilever had several **new compounds** that might work as drugs
- One test = one patient. **Improvement = win, no improvement = loss**
- Testing costs **time and money**, and nobody knows in advance which compound is best
- Unilever asked the mathematician John Gittins: **which compound should we test next, and when should we stop testing?**
- Gittins saw that this is **exactly our casino**: each compound is a slot machine with an unknown win rate

1. For more details see Gittins, J. (1979) "Bandit Processes and Dynamic Allocation Indices", Journal of the Royal Statistical Society. Series B (Methodological).

Gittins discounts the future

- His answer: give every machine a single number, computed only from its own wins and losses, and always play the machine with the highest number. That number is the **Gittins index**.
- Earlier solutions needed a **known interval**
- Gittins instead counts future wins less than immediate ones (**discount factor**), like future cash flows
- **Money:** Money today can earn interest
- **Everyday life:** Tonight's dinner beats the same dinner next week
- So any interval length works, even an endless one

Back to machine A and B

- The Gittins index works for every multi-armed bandit problem
- Let’s consider our two machines one last time:
  - Machine A: 15 pulls, 9 wins, 6 losses
  - Machine B: 2 pulls, 1 win, 1 loss

The Gittins index as a lookup table

Gittins index for different numbers of wins and losses (discount factor 0.9).

LOSSES/WINS	0	1	2	3	4	5	6	7	8	9
0	.7029	.8001	.8452	.8723	.8905	.9039	.9141	.9221	.9287	.9342
1	.5001	.6346	.7072	.7539	.7869	.8115	.8307	.8461	.8588	.8695
2	.3796	.5163	.6010	.6579	.6996	.7318	.7573	.7782	.7956	.8103
3	.3021	.4342	.5184	.5809	.6276	.6642	.6940	.7187	.7396	.7573
4	.2488	.3720	.4561	.5179	.5676	.6071	.6395	.6666	.6899	.7101
5	.2103	.3245	.4058	.4677	.5168	.5581	.5923	.6212	.6461	.6677
6	.1815	.2871	.3647	.4257	.4748	.5156	.5510	.5811	.6071	.6300
7	.1591	.2569	.3308	.3900	.4387	.4795	.5144	.5454	.5723	.5960
8	.1413	.2323	.3025	.3595	.4073	.4479	.4828	.5134	.5409	.5652
9	.1269	.2116	.2784	.3332	.3799	.4200	.4548	.4853	.5125	.5373

Machine A or B by the Gittins index?

- Machine B (0.6346) beats machine A (0.6300)
- Clear **win-stay** pattern, relaxed **lose-shift** pattern
- At (0,0) we see the **exploration bonus**

Gittins Index: the math behind the table

$$G_i(s_i, f_i) := \sup_{\tau \geq 1} \frac{\mathbb{E}\left[\sum_{t=0}^{\tau-1} \beta^t \cdot r_i^t \middle| s_i, f_i\right]}{\mathbb{E}\left[\sum_{t=0}^{\tau-1} \beta^t\right]}$$

In words: imagine a second machine that pays out at a fixed, known rate. The index is the rate that machine would need, so that you'd be indifferent between the two. The more uncertain a machine, the higher its index, because it might turn out great.

> ⓘ **Too complex to calculate by hand**, and you don't need to: just use the lookup table!

Higher index, better choice

- **Core idea:** “What’s the best possible average reward from this option?”
- **Higher index = better choice**
- Less-tested options get an **exploration bonus**, that’s why machine B wins
- The index balances exploration and exploitation automatically

Where bandits run today

- **Online advertising:** which ad to show next, learning from every click
- **Streaming services:** which thumbnail or recommendation to show, tested on millions of viewers
- **E-commerce and A/B testing:** shifting traffic to the better version while the test is still running
- **Clinical trials:** adaptive trials assign more patients to the treatment that works best so far

The same trade-off as in our casino, decided millions of times a day.

Speaker notes

These systems use bandit algorithms in the spirit of Gittins, often simpler variants such as Thompson sampling or UCB. The point for us: explore and exploit is not a thought experiment, it runs the digital economy.

Which rule when?

- Both outcomes known: choose the higher expected value
- E(explore) unknown, interval known: **Bellman** (backward calculation)
- E(explore) unknown, interval unknown: **Gittins index**

Speaker notes

Three situations, three tools. If both outcomes are known, there is nothing to explore. If the interval is known, Bellman’s backward calculation gives the exact answer. If the interval is unknown, the Gittins index does.

Your Personal Action Plan

Question: Rate yourself from 1 (exploit) to 10 (explore):

- **Career:** new skills and roles vs. deepening your expertise
- **Social life:** meeting new people vs. investing in friendships
- **Learning:** new subjects vs. mastering current interests

Time horizon matters: more time left, more exploring.

Speaker notes

Rate yourself honestly. The point is not that exploring is always better: the more time you have left in a phase of your life, the more exploring pays off. Near the end of a phase, exploiting what you know is the better choice.

What we covered

THE DILEMMA: TRY SOMETHING NEW OR STICK WITH WHAT WORKS

Every choice between the new and the known trades information against reward.

TIME DECIDES: WHY THE END OF THE INTERVAL CHANGES EVERYTHING

The more time is left, the more exploring pays off; near the end, exploit what you know.

SMART STRATEGIES: FROM RULES OF THUMB TO THE GITTINS INDEX

Simple rules like win-stay, lose-shift help, but better strategies weigh both what you know and how uncertain you still are.

Why it matters

You have 45 minutes for lunch: the new restaurant or your reliable favorite? Companies face the same choice every day: a new supplier or the proven one, a new marketing channel or the one that works, a new film or another sequel.

Explore early.
Exploit late.

Literature

Interesting literature to start

- Christian, B., & Griffiths, T. (2016). Algorithms to live by: the computer science of human decisions. First international edition. New York, Henry Holt and Company.[1]

1. The main inspiration for this lecture. We have read it and discussed it in depth, always wanting to translate it into a course.

Books on Programming

- Downey, A. B. (2024). Think Python: How to think like a computer scientist (Third edition). O'Reilly. Free online
- Elter, S. (2021). Schrödinger programmiert Python: Das etwas andere Fachbuch (1. Auflage). Rheinwerk Verlag.



Think Python is free online and a great start. Schrödinger programmiert Python is a playful alternative in German.

More Literature

For more interesting literature, take a look at the literature list of this course.